

VStab-QuAD: A New Video-Stabilization Quality Assessment Database

Borhen-eddine Dakkar¹, Amine Bourki², Azeddine Beghdadi¹ and Faouzi Alaya Cheikh³

¹University Sorbonne Paris Nord, France

²VizioSense AI Lab, France

³Norwegian University of Science and Technology (NTNU), Norway

<https://www.l2ti.univ-paris13.fr/databases/>

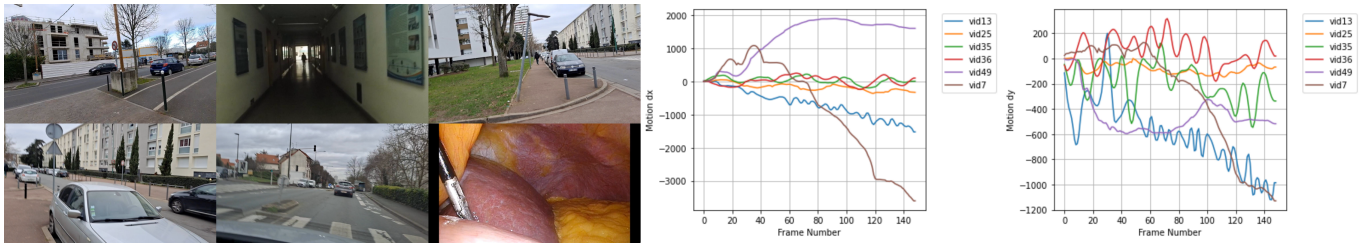


Fig. 1: Overview of the VStab-QuAD database for video stability quality assessment. It includes 320 stabilized video sequences, which stem from five different video stabilization methods that were applied to 64 unstable video sequences associated with various challenging scenarios (e.g., urban, selfies, zoom, medical imagery), image distortions, and perceptual quality scores.

Abstract—We introduce VStab-QuAD, a novel Video-Stabilization Quality Assessment Database, consisting of 320 stabilized video sequences obtained from 64 unstable video sequences with different contents (crowd, parallax, running, zooming, etc) using five different strong performing video-stabilization methods. The stabilized videos that are considered contain residual instability-induced image distortions and additional distortions introduced as a side effect of the digital stabilization algorithms applied. In addition, this study presents a comprehensive objective analysis of the video sequences and their validity for video-stabilization quality assessment. Subjective experiments were performed in a controlled environment using the single stimulus continuous quality evaluation (SSCQE) protocol, and their results are included in the database. Therefore, our proposed VStab-QuAD contains (i) the 64 original videos with in-capture distortions due to initial video instability, (ii) the 320 stabilized videos produced with the five different algorithms, and (iii) associated perceptual quality scores.

This paper hence contributes VStab-QuAD as a comprehensive public benchmark database designed to facilitate the development of powerful video-stabilization methods and quality assessment metrics in real-life video acquisition scenarios.

Index Terms—Video Stabilization, Subjective Quality Assessment, Image Distortions, Datasets and Evaluation, Video Processing.

I. INTRODUCTION

Thanks to the growing number of devices equipped with cameras, the production of videos has registered a notable increase [1]. In 2020, there were over three billion internet users who watched or downloaded a video content [1]. Most of the hand-held devices make it easy to capture video. The captured video with such mobile devices, are generally shaky and suffer from unwanted motions. As the human visual

system is very sensitive to distortions in such visual material, the instability of a video is perceived as a disturbing visual degradation affecting the view experience.

This made video-stabilization one of the most important applications of video processing. It aims to improve the visual quality of video sequences by eliminating the unwanted motion. A video stabilization method generally consists of two major steps: motion estimation step and motion compensation step. Motion estimation allows the estimation of the global camera motion vectors. While in the motion compensation, global motion vectors are compensated for to remove the frames jitter and to produce smooth motion trajectories.

A robust video stabilization method should satisfy three major criteria regarding the perceptual quality: stability, cropping and visual distortion [2]. An unstable video means the presence of object instabilities in the whole frame. Cropping is characterised by irregular boundaries after warping frames. As the warped frames are cropped to obtain a rectangular boundary for normal video display. The visual distortion criteria could be any perceptual visual discomfort, e.g, blur, noise, objects geometry or illumination.

Most of the existing datasets are destined to video-stabilization methods development and testing [3]. To the best of our knowledge, no dataset were specifically dedicated to video-stabilization quality assessment. Our VStab-QuAD is the first one to fill that space. It aims to facilitate the quantification and benchmarking of video-stabilization algorithms. A new subjective methodology is proposed, that takes into account the complexity of the perceptual video stabilization problems.

The main contributions of this work are:

- VStab-QuAD, a novel database containing multiple challenging scenarios that can be used to assess the performance of video stabilization methods.
- A subjective video quality study using a new protocol for the evaluation of the distortions that are typically the by-product of digital video-stabilization.
- A new subjective quality score measure using a fusion scheme based on human observers' preferences. It is expressed as an explicit combination of the visual- and stability quality scores.

II. RELATED WORK

Video Stabilization Quality Assessment (VSQA) methods can be divided into two main categories based on subjective, and objective assessment strategies.

Subjective assessment involves the investigation of the quality by asking people's judgement. A final score representing the quality is given for each sequence. The authors of [4] proposed a comparison of stabilized versions of synthetically distorted endoscopic videos and their synthetic full-reference ones. They have generated synthetically unstable videos. Then, four video-stabilization methods have been used to stabilize the generated videos. The quality were determined by comparing the videos to their respective full-reference ones. This dataset was limited only to the endoscopic Videos. In [5], a dataset of stable and shaky videos was constructed by capturing realistic stable/shaky videos. Then, digital video-stabilization algorithms have been running on shaky videos to obtain the stabilized sequences. They have used the Double Stimulus Continuous Quality Scale (DSCQS) to assess the subjective scores. A mean opinion score (MOS) is associated to each video (ranging from 1, being perfectly stable, to 5, extremely shaky). The dataset assesses only the stability and does not take into account other stabilization distortions. In [6], they propose a user study based on MOS to measure the subject preference based on the paired comparison between different methods. Observers are shown several videos and are asked to select the best one according to some predefined metrics. Zhang et al. [7] proposed a dataset composed of shaky videos and their corresponding stabilized ones by running some classic video stabilization methods. The stability is ranked by choosing the best stable video. The average rank is computed for each video to give the final order of the subjective judgement.

All the state-of-the-art proposed datasets evaluate only the stability [2], [8]–[10]. However, video-stabilization methods introduce other degradations that affect the perceptual quality. Neglecting these degradations gives a biased assessment. In this work, we intend to investigate all the video-stabilization impairments.

III. THE VSTAB-QUAD DATABASE

We have constructed a new Video-Stabilization Quality Assessment (VStab-QuAD) dataset, and then conducted a

human subjective study. A new protocol dedicated to the video-stabilization QA has been adopted.

A. Acquisition of the Initial Unstable Videos

There are a total of 64 videos in different categories, including selfie, rotation, zooming, running, climbing, driving, endoscopic, rolling shutter, dark, and crowd with large parallax, among others. These videos are provided in raw AVI format, with frame rates ranging from 24 to 30 frames per second, and durations ranging from 5 to 10 seconds. Out of the 64 videos, 46 are novel acquisitions, and 18 were obtained from related to existing video-stabilization method papers [11]–[14].

B. Generation of Stabilized Videos

To produce stabilization versions of the initially unprocessed, real, video data, we consider 5 well established methods. The 5 softwares are (i) Adobe After Effects (Ae) warp stabilizer, (ii) Google Photos application, (iii) VirtualDub Deshaker, (iv) vReveal and (v) VideoProc Converter [15].

AE stabilizer is based on the subspace method in [4], and YouTube implements the method of L1-optimization on inter-frame transformations [2]. 320 stabilized videos are then produced by using these methods leading to a very broad range of motions trajectories and novel visual artefacts that typically stem from video stabilization (Figure 2).

C. Dataset Characteristics Analysis

It is important to have different scenarios but also visual contents rich in spatio-temporal structures at different scales of observation and under various lighting and viewing angles. The richness or diversity in videos is computed through spatial and spatio-temporal descriptors. A set of criteria and measures to quantify and analyze the richness and representativeness of image and video databases dedicated to perceptual quality assessment has been proposed in [16], [17]. In what follows we will recall the Saptio-temporal Descriptors and apply them to the analysis of the database.

D. Saptio-temporal Descriptors

The richness of a video signal in terms of visual time-varying content is measured using the Spatio-temporal information. The VQEG group formula has been used [18].

1) *Spatial perceptual information*: We have used the Sobel operator to enhance the edges as in [18]. We applied a Sobel filter for each video frame (F_n). The standard deviation of the magnitude of the Sobel response is then computed. The maximum value over time is used to represent the spatial information content of the video sequence.

$$SI = \max_n \{ \text{std}[\text{Sobel}(F_n)] \} \quad (1)$$

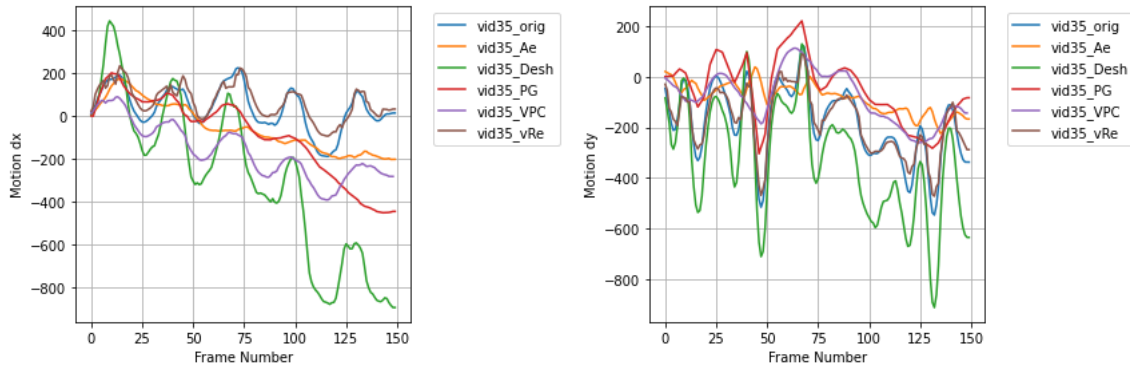


Fig. 2: An example video with its associated motion trajectory according to its horizontal (dx) and vertical (dy) components. The original trajectory is in blue, and the 5 video stabilization methods we consider, present the remaining colors.

2) *Temporal perceptual information*: To capture the motion difference at a location (i, j) , the temporal perceptual information denoted $M_n(i, j)$ is used. It is computed as follows:

$$M_n(i, j) = F_n(i, j) - F_{n-1}(i, j) \quad (2)$$

where $F_n(i, j)$ is the pixel value at the (i, j) location of n^{th} frame. The TI measure is computed as the maximum over time (n) of the standard deviation over space of inter-frame difference M_n :

$$TI = \max_n \{\text{std}[M_n]\} \quad (3)$$

High motion Videos are characterized with high TI values. Figure 3 presents a scatter plot of the spatio-temporal descriptors SI and TI of the original shaky videos. It can be seen that the still scenes and those with very limited motion are found when SI is close to 0 (for samples 18 and 14), while scenes with a lot of motion are found near the upper part of the plot (for 63, 28, 32). Scenes with minimal spatial detail are located on the left side of the plot (18, 14, 19), while scenes with the most spatial detail are located on the right side of the plot (12, 32, 28).

E. Motion trajectory

Optical flow has been used to estimate the motion trajectory in the original videos. First, features are extracted from each frame and tracked using the Lucas-Kanade Optical Flow algorithm. The motion is then estimated using a rigid Euclidean transformation. The motion trajectory of some original videos in the x and y directions are presented in Figure 1.

IV. SUBJECTIVE STUDY FOR A JOINT VISUAL AND STABILIZATION QA

A. Experimental Setup

The testing environment included a i7-7700 CPU @ 3.60GHz PC equipped with 1920×1080 monitor. A user interface was designed whereby the subjects could view and rate the videos. The observers begin the test by registering their information. In the next tab, the test instructions are explained. Two sessions are proposed in the following: Train session and

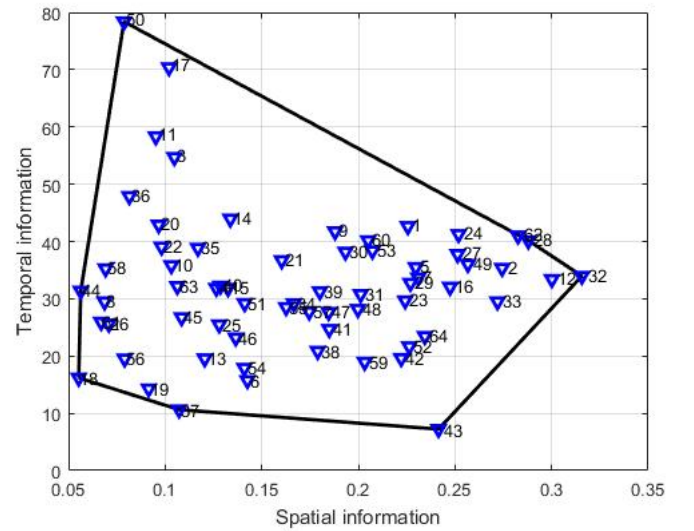


Fig. 3: Spatio-temporal diversity of the VStab-QuAD videos by means of their perceptual Spatial (SI) and Temporal (TI) criteria

Test session respectively. The aim of the Train session is to familiarize the observers with the test expectations. The Test session contains the real test where each subjective score is recorded. At the end, the observers are asked to give their preferences between the visual quality and the stabilization quality.

B. Experimental Protocol

Most of the existing video-stabilization metrics or datasets only consider stabilization-related evaluation criteria. However, other degradations may appear as a by-product of video stabilization. In order to obtain representative data on how humans perceive the video-stabilization distortions, we have conducted a subjective study to obtain subjective scores. For this, we have adopted a new protocol based on single stimulus continuous quality evaluation (SSCQE) to obtain the subjective

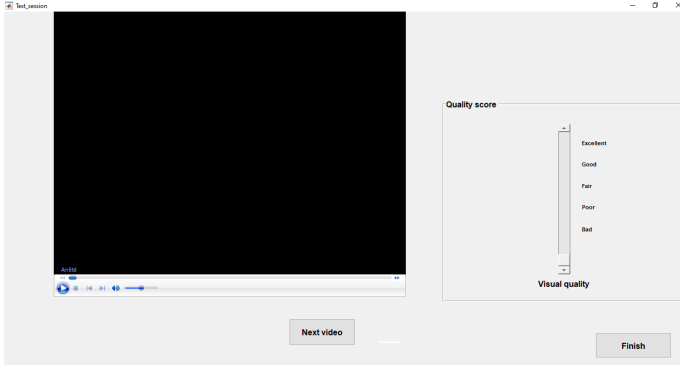


Fig. 4: GUI of our SSCQE protocol to construct our subjective quality ratings.

quality scores. It rates all of the stabilized videos. As we have reported before, evaluating a video-stabilization method regarding only its stability is not sufficient. In this study, the overall quality score is obtained by evaluating two criteria, the visual quality and the stabilization quality.

1) **Visual quality:** The removal of instabilities is often accompanied by the introduction of some side effects that affect the visual quality of the video sequences. In this part of the study, the observers are asked to evaluate the perceptual quality with regard to the following distortions:

- **Blur:** it is characterized by the lack of sharpness.
- **Twisted backgrounds:** this distortion mainly occurs on backgrounds. It represents a wave-like movement on the background.
- **Vibration (shudder):** it appears as tremors in the sequences.

2) **Stabilization quality:** The main objective of digital video stabilization is the improvement of the video quality by removing unwanted camera motion. Two scenarios are possible after the application of a such video-stabilization method:

- i- The video remains unstable. It means that the stabilization process is inefficient with regard to the jitters presented in the video.
- ii- The moving objects present some latency. This distortion is the consequence of the suppression of the wanted motion in the video.

The experiments are performed to evaluate the visual quality and the stabilization quality. A continuous quality slider is used to derive the subjective scores. The quality slider is labeled with five adjectives as mentioned in Table I. All subjects were instructed to give an opinion quality score of the overall perceived video sequences. They were seated to perform the experiments at a fixed distance of twice the screen height. While this contrasts slightly with our overall experimental setup which mainly follows [18], we have decided to consider this typical distance relative to the considered screen resolution in order to induce more critical disparities within the gathered

TABLE I: Adjectival categorical judgement

Stabilization quality	Visual quality
Very annoying	Bad
Annoying	Poor
Slightly annoying	Fair
Perceptible, but not annoying	Good
Imperceptible	Excellent

ratings by allowing the observers to assess more clearly the distortions introduced by the stabilization process. This viewing distance was maintained during each test session, and throughout the considered observers. All the observers had either normal vision or corrected to normal vision.

A total of 15 subjects participated in the study, most of them are graduate students. The subjective test is started with general information about the study and instructions on how to participate to the video-stabilization task. In a short training session, 10 videos were played to familiarize subjects with the user interface. Following this, a test session was initiated. 160 stabilized videos were randomly and equally divided into 2 sessions. The subjects participated in 4 sessions, 2 for visual quality and 2 for stabilization quality. These sessions were separated by at least 24 hours. After completion of both testing sessions, subjects were asked to indicate their preference between stability and visual quality. This information will be used as a weighted trade-off parameter for the combination score.

C. Joint Visual and Stabilization Subjective Scores

The subjective score is computed as indicated in the ITU-R recommendation [19]. The ratings was converted into mean opinion scores (MOS).

$$MOS_j = \frac{1}{N} \sum_{i=1}^N r_{ij} \quad (4)$$

where r_{ij} represents the ratings of the j th image given by the i th subject and N is the number of subjects.

We named $MOSV$ and $MOSS$ the obtained score from the visual quality test and the obtained score from the stabilization quality test respectively.

The overall score of one video $MOSF$ is obtained using a the preference weight as follows:

$$MOSF_j = \alpha MOSV_j + (1 - \alpha) MOSS_j \quad (5)$$

where α is a tuning parameter which encodes the relative preference between the perceptual visual quality and stabilization quality. It is inferred by means of a subjective survey as detailed further in the next sub-section.

D. Subjective scores

Figure 5 displays the histogram of Mean Opinion Scores (MOSs) across the entire database. The plot shows a wide range of perceptual quality scores, spanning all possible values. This indicates that the dataset contains a rich variety of MOS levels throughout.

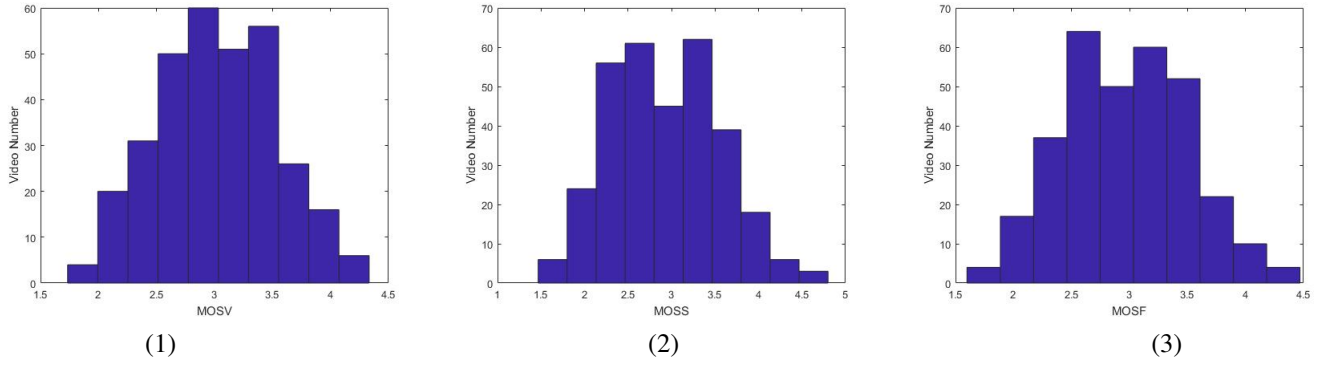


Fig. 5: **Histogram of MOS scores covering the VStab-QA.** This includes MOSV for visual quality (1), MOSS for stabilization quality (2), and MOSF for the final joint MOS (3).

TABLE II: Relative preference weight between Visual and Stabilization quality, as estimated by surveying a human cohort.

Survey size	Visual quality	Stabilization quality
15	33.33%	66.66%

Table II presents the results of the subject preferences. We can notice that 66% of the subjects prefer a stable video, while 33% prefer a better visual quality. Regarding the preference results, we have fixed the preference weight to $\alpha = 0.33$. This parameter has been used to compute the overall MOS score.

V. CONCLUSION AND PERSPECTIVES

We have introduced VStab-QuAD, a novel database dedicated to quality assessment of video stabilization methods. It is comprised of 320 stabilized video sequences from five different video stabilization methods that were applied to 64 unstable video sequences associated with various challenging scenarios, along visual- and stability focused MOSs. We proposed an experimental study using subjective scoring, as well as a novel combined score that jointly considers perceptual visual quality and video stabilization quality in a structured, unbiased way. In particular, our combined score will pave the way for the ever-growing body of literature on video stabilization in a blind, reference-free context [20]–[22] while taking into account their impact on visual quality.

Future work includes studying the correlation of objective metrics with our subjective unified MOS, which encompasses the considerations of stabilization and visual quality, to help cope with the by-product artifacts that are typically generated by VS methods.

REFERENCES

- [1] “statista,” <https://www.statista.com/statistics/1061017/digital-video-viewers-number-worldwide/>, 2023.
- [2] S. Liu, L. Yuan, P. Tan, and J. Sun, “Bundled camera paths for video stabilization,” *ACM Trans. Graph.*, vol. 32, no. 4, jul 2013. [Online]. Available: <https://doi.org/10.1145/2461912.2461995>
- [3] Y. Wang, Q. Huang, C. Jiang, J. Liu, M. Shang, and Z. Miao, “Video stabilization: A comprehensive survey,” *Neurocomput.*, vol. 516, no. C, p. 205–230, dec 2022. [Online]. Available: <https://doi.org/10.1016/j.neucom.2022.10.008>
- [4] M. C. Offiah, N. Amin, T. Gross, N. El-Sourani, and M. Borschbach, “An approach towards a full-reference-based benchmarking for quality-optimized endoscopic video stabilization systems,” in *Proceedings of the Eighth Indian Conference on Computer Vision, Graphics and Image Processing*, ser. ICVGIP ’12. New York, NY, USA: Association for Computing Machinery, 2012. [Online]. Available: <https://doi.org/10.1145/2425333.2425398>
- [5] L. Zhang, Q.-Z. Zheng, H.-K. Liu, and H. Huang, “Full-reference stability assessment of digital video stabilization based on riemannian metric,” *IEEE Transactions on Image Processing*, vol. 27, no. 12, pp. 6051–6063, 2018.
- [6] W. Guilluy, A. Beghdadi, and L. Oudre, “A performance evaluation framework for video stabilization methods,” in *2018 7th European Workshop on Visual Information Processing (EUVIP)*, 2018, pp. 1–6.
- [7] L. Zhang, Q.-Z. Zheng, and H. Huang, “Intrinsic motion stability assessment for video stabilization,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 4, pp. 1681–1692, 2019.
- [8] A. Goldstein and R. Fattal, “Video stabilization using epipolar geometry,” *ACM Transactions on Graphics (TOG)*, vol. 31, no. 5, pp. 1–10, 2012.
- [9] Y. J. Koh, C. Lee, and C.-S. Kim, “Video stabilization based on feature trajectory augmentation and selection and robust mesh grid warping,” *IEEE Transactions on Image Processing*, vol. 24, no. 12, pp. 5260–5273, 2015.
- [10] M. Wang, G.-Y. Yang, J.-K. Lin, S.-H. Zhang, A. Shamir, S.-P. Lu, and S.-M. Hu, “Deep online video stabilization with multi-grid warping transformation learning,” *IEEE Transactions on Image Processing*, vol. 28, no. 5, pp. 2283–2292, 2018.
- [11] S. Liu, L. Yuan, P. Tan, and J. Sun, “Bundled camera paths for video stabilization,” *ACM Trans. Graph.*, vol. 32, no. 4, jul 2013. [Online]. Available: <https://doi.org/10.1145/2461912.2461995>
- [12] J. Yu, R. Ramamoorthi, K. Cheng, M. Sarkis, and N. Bi, “Real-time selfie video stabilization,” in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 12 031–12 039.
- [13] Z. Shi, F. Shi, W.-S. Lai, C.-K. Liang, and Y. Liang, “Deep online fused video stabilization,” in *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022, pp. 865–873.
- [14] A. Beghdadi, M. A. Qureshi, B.-E. Dakkar, H. H. Gillani, Z. A. Khan, M. Kaaniche, M. Ullah, and F. A. Cheikh, “A new video quality assessment dataset for video surveillance applications,” in *2022 IEEE International Conference on Image Processing (ICIP)*, 2022, pp. 1521–1525.
- [15] “Videoproc,” <https://www.videoproc.com>, 2023.
- [16] S. Winkler, “Analysis of public image and video databases for quality assessment,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 6, no. 6, pp. 616–625, 2012.

- [17] A. Beghdadi, M. A. Qureshi, B. Sdiri, M. Deriche, and F. Alaya-Cheikh, "Ceed-a database for image contrast enhancement evaluation," in *Colour and Visual Computing Symposium (CVCS)*. IEEE, 2018, pp. 1–6.
- [18] P. ITU-T RECOMMENDATION, "Subjective video quality assessment methods for multimedia applications," *International telecommunication union*, 1999.
- [19] B. Series, "Methodology for the subjective assessment of the quality of television pictures," *Recommendation ITU-R BT*, vol. 500, pp. 500–13, 2012.
- [20] J. Yu, R. Ramamoorthi, K. Cheng, M. Sarkis, and N. Bi, "Real-time selfie video stabilization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 12 036–12 044.
- [21] A. Ciancio, E. A. da Silva, A. Said, R. Samadani, P. Obrador, et al., "No-reference blur assessment of digital pictures based on multifeature classifiers," *IEEE Transactions on image processing*, vol. 20, no. 1, pp. 64–75, 2010.
- [22] Y. Fang, H. Zhu, Y. Zeng, K. Ma, and Z. Wang, "Perceptual quality assessment of smartphone photography," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3677–3686.
- [23] D. Li, T. Jiang, and M. Jiang, "Quality assessment of in-the-wild videos," in *27th ACM International Conference on Multimedia*, 2019, pp. 2351–2359.
- [24] F. Götz-Hahn, V. Hosu, H. Lin, and D. Saupe, "Konvid-150k: A dataset for no-reference video quality assessment of videos in-the-wild," *IEEE Access*, vol. 9, pp. 72 139–72 160, 2021.
- [25] M. Nuutinen, T. Virtanen, M. Vaahteranoksa, T. Vuori, P. Oittinen, and J. Häkkinen, "Cvd2014—a database for evaluating no-reference video quality assessment algorithms," *IEEE Transactions on Image Processing*, vol. 25, no. 7, pp. 3073–3086, 2016.
- [26] V. Hosu, F. Hahn, M. Jenadeleh, H. Lin, H. Men, T. Szirányi, S. Li, and D. Saupe, "The konstanz natural video database (konvid-1k)," in *9th International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, 2017, pp. 1–6.
- [27] D. Ghadiyaram, J. Pan, A. C. Bovik, A. K. Moorthy, P. Panda, and K.-C. Yang, "In-capture mobile video distortions: A study of subjective behavior and objective algorithms," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 28, no. 9, pp. 2061–2077, 2017.
- [28] Z. Sinno and A. C. Bovik, "Large-scale study of perceptual video quality," *IEEE Transactions on Image Processing*, vol. 28, no. 2, pp. 612–627, 2018.
- [29] R. Kumar, H. Sawhney, S. Samarasekera, S. Hsu, H. Tao, Y. Guo, K. Hanna, A. Pope, R. Wildes, D. Hirvonen, et al., "Aerial video surveillance and exploitation," *Proceedings of the IEEE*, vol. 89, no. 10, pp. 1518–1539, 2001.
- [30] I. Bezzine, Z. A. Khan, A. Beghdadi, N. Al-Maadeed, M. Kaaniche, S. Al-Maadeed, A. Bouridane, and F. A. Cheikh, "Video quality assessment dataset for smart public security systems," in *23rd International Multipoint Conference (INMIC)*. IEEE, 2020, pp. 1–5.
- [31] Z. A. Khan, A. Beghdadi, F. A. Cheikh, M. Kaaniche, E. Pelanis, R. Palomar, Å. A. Fretland, B. Edwin, and O. J. Elle, "Towards a video quality assessment based framework for enhancement of laparoscopic videos," in *Medical Imaging: Image Perception, Observer Performance, and Technology Assessment*, vol. 11316. International Society for Optics and Photonics, 2020, p. 113160P.
- [32] F. De Simone, M. Naccari, M. Tagliasacchi, F. Dufaux, S. Tubaro, and T. Ebrahimi, "Subjective assessment of h. 264/avc video sequences transmitted over a noisy channel," in *International Workshop on Quality of Multimedia Experience*. IEEE, 2009, pp. 204–209.
- [33] F. Zhang, S. Li, L. Ma, Y. C. Wong, and K. N. Ngan, "Ivp subjective quality video database," The Chinese University of Hong Kong, <http://ivp.ee.cuhk.edu.hk/research/database/subjective>, 2011.
- [34] M. A. Qureshi, A. Beghdadi, and M. Deriche, "Towards the design of a consistent image contrast enhancement evaluation measure," *Signal Processing: Image Communication*, vol. 58, pp. 212–227, 2017.
- [35] Z. Mortezaie, H. Hassanpour, and A. Beghdadi, "People re-identification under occlusion and crowded background," *Multimedia Tools and Applications*, pp. 1–21, 2022.
- [36] P. Bouttefroy, A. Bouzerdoum, S. Phung, and A. Beghdadi, "Abnormal behavior detection using a multi-modal stochastic learning approach," in *2008 International Conference on Intelligent Sensors, Sensor Networks and Information Processing*. IEEE, 2008, pp. 121–126.
- [37] P. L. M. Bouttefroy, A. Bouzerdoum, S. L. Phung, and A. Beghdadi, "Vehicle tracking using projective particle filter," in *2009 Sixth IEEE International Conference on Advanced Video and Signal Based Surveillance*. IEEE, 2009, pp. 7–12.
- [38] D. Hasler and S. E. Suesstrunk, "Measuring colorfulness in natural images," in *Human vision and electronic imaging VIII*, vol. 5007. International Society for Optics and Photonics, 2003, pp. 87–95.
- [39] H. Yu and S. Winkler, "Image complexity and spatial information," in *5th International Workshop on Quality of Multimedia Experience (QoMEX)*. IEEE, 2013, pp. 12–17.
- [40] K. Matkovic, L. Neumann, A. Neumann, T. Psik, and W. Purghofer, "Global contrast factor-a new approach to image contrast," *Computational Aesthetics*, vol. 2005, no. 159-168, p. 1, 2005.
- [41] K. Garg and S. K. Nayar, "When does a camera see rain?" in *10th International Conference on Computer Vision (ICCV)*. IEEE, 2005, pp. 1067–1074.
- [42] A. Beghdadi, M. Asim, N. Almaadeed, and M. A. Qureshi, "Towards the design of smart video-surveillance system," in *NASA/ESA Conference on Adaptive Hardware and Systems (AHS)*. IEEE, 2018, pp. 162–167.
- [43] A. Beghdadi, I. Bezzine, and M. A. Qureshi, "A perceptual quality-driven video surveillance system," in *23rd International Multipoint Conference (INMIC)*. IEEE, 2020, pp. 1–6.
- [44] A. Beghdadi, M. A. Qureshi, S. A. Amirshahi, A. Chetouani, and M. Pedersen, "A critical analysis on perceptual contrast and its use in visual information analysis and processing," *IEEE Access*, vol. 8, pp. 156 929–156 953, 2020.
- [45] E. Kalalembang, K. Usman, and I. P. Gunawan, "Dct-based local motion blur detection," in *International Conference on Instrumentation, Communication, Information Technology, and Biomedical Engineering*. IEEE, 2009, pp. 1–6.
- [46] X. Min, G. Zhai, J. Zhou, M. C. Q. Farias, and A. C. Bovik, "Study of subjective and objective quality assessment of audio-visual signals," *IEEE Transactions on Image Processing*, vol. 29, pp. 6054–6068, 2020.
- [47] J. Choi, J. Park, and I. S. Kweon, "Self-supervised real-time video stabilization," *arXiv preprint arXiv:2111.05980*, 2021.
- [48] M. K. Ali, S. Yu, and T. H. Kim, "Deep motion blind video stabilization," *arXiv preprint arXiv:2011.09697*, 2020.
- [49] Z. Shi, F. Shi, W.-S. Lai, C.-K. Liang, and Y. Liang, "Deep online fused video stabilization," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 1250–1258.
- [50] Y.-T. Chen, K.-W. Tseng, Y.-C. Lee, C.-Y. Chen, and Y.-P. Hung, "Pixstabenet: Fast multi-scale deep online video stabilization with pixel-based warping," in *2021 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2021, pp. 1929–1933.
- [51] Y.-L. Liu, W.-S. Lai, M.-H. Yang, Y.-Y. Chuang, and J.-B. Huang, "Hybrid neural fusion for full-frame video stabilization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 2299–2308.
- [52] M. Zhao and Q. Ling, "Pwstabenet: Learning pixel-wise warping maps for video stabilization," *IEEE Transactions on Image Processing*, vol. 29, pp. 3582–3595, 2020.
- [53] S. Liu, P. Tan, L. Yuan, J. Sun, and B. Zeng, "Meshflow: Minimum latency online video stabilization," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part VI 14*. Springer, 2016, pp. 800–815.
- [54] L. Zhang, Q.-Z. Zheng, H.-K. Liu, and H. Huang, "Full-reference stability assessment of digital video stabilization based on riemannian metric," *IEEE Transactions on Image Processing*, vol. 27, no. 12, pp. 6051–6063, 2018.
- [55] Q. Rao, X. Yu, S. Navasardyan, and H. Shi, "Sim2realvs: A new benchmark for video stabilization with a strong baseline," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 5406–5415.
- [56] M. Sarigül, "A survey on digital video stabilization," *Multimedia Tools and Applications*, pp. 1–27, 2023.